1 The constrained proximal subproblem

For a proper closed convex function h and $\gamma > 0$, define its proximal and Moreau envelope operators by

$$\operatorname{prox}_{\gamma h}(\boldsymbol{v}) := \arg\min_{\boldsymbol{x}} \left\{ h(\boldsymbol{x}) + \frac{1}{2\gamma}\|\boldsymbol{x}-\boldsymbol{v}\|_2^2 \right\}, \qquad e_{\gamma h}(\boldsymbol{v}) := \min_{\boldsymbol{x}} \left\{ h(\boldsymbol{x}) + \frac{1}{2\gamma}\|\boldsymbol{x}-\boldsymbol{v}\|_2^2 \right\}.$$

Let $\mathcal{C} := \{\boldsymbol{x} \in \mathbb{R}^d : \boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}\}$ and $\widetilde{G} := G + \delta_{\mathcal{C}}$. The proximal subproblem of interest is

$$\boldsymbol{x}^\star := \operatorname{prox}_{\gamma \widetilde{G}}(\boldsymbol{v}) = \arg\min_{\boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}} \left\{ p_{\boldsymbol{v}}(\boldsymbol{x}) := G(\boldsymbol{x}) + \frac{1}{2\gamma}\|\boldsymbol{x}-\boldsymbol{v}\|_2^2 \right\}, \qquad e_{\gamma \widetilde{G}}(\boldsymbol{v}) = p_{\boldsymbol{v}}(\boldsymbol{x}^\star).$$

Assume that G is proper, closed, convex, and piecewise linear–quadratic, that $\operatorname{prox}_{\gamma G}$ is inexpensive, and that a Slater point exists. For the numerical experiment below, the base oracle is the PAVA oracle from our paper for

$$G(\boldsymbol{x}) := \min_{\boldsymbol{z}} \left\{ \frac{1}{2}\sum_{i=1}^{d} \frac{x_i^2}{z_i} : \boldsymbol{0} \leq \boldsymbol{z} \leq \boldsymbol{1},\ \boldsymbol{1}^\top \boldsymbol{z} \leq k,\ -Mz_i \leq x_i \leq Mz_i \right\}.$$

1.1 Primal–dual reduction

For $\boldsymbol{\lambda} \geq \boldsymbol{0}$, minimizing the Lagrangian over $\boldsymbol{x}$ gives

$$\begin{aligned}
\boldsymbol{x}(\boldsymbol{\lambda}) &:= \operatorname{prox}_{\gamma G}\left(\boldsymbol{v} - \gamma \boldsymbol{A}^\top \boldsymbol{\lambda}\right), \\
q_{\boldsymbol{v}}(\boldsymbol{\lambda}) &:= e_{\gamma G}\left(\boldsymbol{v} - \gamma \boldsymbol{A}^\top \boldsymbol{\lambda}\right) + \boldsymbol{\lambda}^\top(\boldsymbol{A}\boldsymbol{v} - \boldsymbol{b}) - \frac{\gamma}{2}\|\boldsymbol{A}^\top \boldsymbol{\lambda}\|_2^2, \\
e_{\gamma \widetilde{G}}(\boldsymbol{v}) &= \max_{\boldsymbol{\lambda} \geq \boldsymbol{0}} q_{\boldsymbol{v}}(\boldsymbol{\lambda}), \qquad \nabla q_{\boldsymbol{v}}(\boldsymbol{\lambda}) = \boldsymbol{A}\boldsymbol{x}(\boldsymbol{\lambda}) - \boldsymbol{b}, \qquad L_{\mathrm{d}} := \gamma\|\boldsymbol{A}\|_2^2.
\end{aligned}$$

Thus, both algorithms below use only $\operatorname{prox}_{\gamma G}$ and products with $\boldsymbol{A}$ and $\boldsymbol{A}^\top$. For a computable stopping certificate, take a strict anchor $\bar{\boldsymbol{x}}$ with $\boldsymbol{A}\bar{\boldsymbol{x}} < \boldsymbol{b}$, and define

$$\boldsymbol{s} := \boldsymbol{b} - \boldsymbol{A}\bar{\boldsymbol{x}}, \qquad \boldsymbol{u} := [\boldsymbol{A}\boldsymbol{x}(\boldsymbol{\lambda}) - \boldsymbol{b}]_+, \qquad \theta := \max_i \frac{u_i}{u_i + s_i}, \qquad \widehat{\boldsymbol{x}} := (1-\theta)\boldsymbol{x}(\boldsymbol{\lambda}) + \theta\bar{\boldsymbol{x}},$$

$$\Delta(\widehat{\boldsymbol{x}}, \boldsymbol{\lambda}) := p_{\boldsymbol{v}}(\widehat{\boldsymbol{x}}) - q_{\boldsymbol{v}}(\boldsymbol{\lambda}) \geq 0, \qquad \|\widehat{\boldsymbol{x}} - \boldsymbol{x}^\star\|_2 \leq \sqrt{2\gamma\Delta(\widehat{\boldsymbol{x}}, \boldsymbol{\lambda})}.$$

Hence, $\Delta \leq \varepsilon$ certifies objective accuracy ε, while $\Delta \leq \varepsilon^2/(2\gamma)$ certifies proximal-point accuracy ε.

2 Two inner solvers

Algorithm 1 Restarted dual FISTA

Input: $\boldsymbol{v}, \gamma, L_{\mathrm{d}}$, restart length T, and tolerance ε.

Set $\boldsymbol{\lambda} = \boldsymbol{y} = \boldsymbol{0}$ and $t = 1$.
for $j = 0, 1, \ldots$ **do**
  $\boldsymbol{x}_{\boldsymbol{y}} = \operatorname{prox}_{\gamma G}(\boldsymbol{v} - \gamma \boldsymbol{A}^\top \boldsymbol{y})$ and $\boldsymbol{\lambda}^+ = [\boldsymbol{y} + L_{\mathrm{d}}^{-1}(\boldsymbol{A}\boldsymbol{x}_{\boldsymbol{y}} - \boldsymbol{b})]_+$.
  $t^+ = (1 + \sqrt{1 + 4t^2})/2$ and $\boldsymbol{y}^+ = \boldsymbol{\lambda}^+ + (t-1)(\boldsymbol{\lambda}^+ - \boldsymbol{\lambda})/t^+$.
  Set $(\boldsymbol{\lambda}, \boldsymbol{y}, t) = (\boldsymbol{\lambda}^+, \boldsymbol{y}^+, t^+)$.
  if $j+1$ is a multiple of T **then**
    Restart with $\boldsymbol{y} = \boldsymbol{\lambda}$ and $t = 1$; compute $\widehat{\boldsymbol{x}}$ and $\Delta(\widehat{\boldsymbol{x}}, \boldsymbol{\lambda})$.
    if $\Delta(\widehat{\boldsymbol{x}}, \boldsymbol{\lambda}) \leq \varepsilon$ **then**
      return $\widehat{\boldsymbol{x}}$.
    end if
  end if
end for

Let $\psi_{\boldsymbol{v}} := -q_{\boldsymbol{v}} + \delta_{\mathbb{R}_+^m}$, $\boldsymbol{\Lambda}^\star := \arg\min \psi_{\boldsymbol{v}}$, and $D(\boldsymbol{\lambda}) := \psi_{\boldsymbol{v}}(\boldsymbol{\lambda}) - \min \psi_{\boldsymbol{v}}$. Since G is PLQ, $\psi_{\boldsymbol{v}}$ is convex PLQ. If the initial dual sublevel set is bounded, then for some $\mu_{\boldsymbol{v}} > 0$ it satisfies $D(\boldsymbol{\lambda}) \geq \mu_{\boldsymbol{v}} \operatorname{dist}^2(\boldsymbol{\lambda}, \boldsymbol{\Lambda}^\star)/2$ on that set. One exact FISTA epoch therefore satisfies

$$D(\boldsymbol{\lambda}_{s+1}) \leq \frac{4L_{\mathrm{d}}}{\mu_{\boldsymbol{v}}(T+1)^2} D(\boldsymbol{\lambda}_s).$$

Taking $T + 1 \geq \sqrt{8L_{\mathrm{d}}/\mu_{\boldsymbol{v}}}$ halves the error at every restart and gives the following complexity for proximal-point accuracy ε:

$$N_{\text{FISTA}} = \mathcal{O}\left(\sqrt{\frac{L_{\mathrm{d}}}{\mu_{\boldsymbol{v}}}} \log \frac{C_{\boldsymbol{v}}}{\varepsilon}\right)$$

Here, the bound follows from $\|\boldsymbol{x}(\boldsymbol{\lambda}) - \boldsymbol{x}^\star\|_2^2 \le 2\gamma D(\boldsymbol{\lambda})$. The rate is linear at the restart points; $\mu_{\boldsymbol{v}}$ is instance dependent and usually unknown, so the implementation treats T as a tunable parameter, while the feasible certificate has the same order under the usual local Lipschitz recovery bound.

For Newton, define the projected KKT residual

$$\mathcal{R}_{\boldsymbol{v}}(\boldsymbol{\lambda}) := \boldsymbol{\lambda} - [\boldsymbol{\lambda} + c(\boldsymbol{A}\boldsymbol{x}(\boldsymbol{\lambda}) - \boldsymbol{b})]_+.$$

Choose $\boldsymbol{J} \in \partial_{\mathrm{B}} \operatorname{prox}_{\gamma G}(\boldsymbol{v} - \gamma \boldsymbol{A}^\top \boldsymbol{\lambda})$ from the current PAVA pools, and let $\boldsymbol{E}$ be the diagonal derivative of the positive-part map at $\boldsymbol{\lambda} + c(\boldsymbol{A}\boldsymbol{x}(\boldsymbol{\lambda}) - \boldsymbol{b})$. Then

$$\boldsymbol{V} := \boldsymbol{I} - \boldsymbol{E} + c\gamma \boldsymbol{E}\boldsymbol{A}\boldsymbol{J}\boldsymbol{A}^\top \in \partial_{\mathrm{B}} \mathcal{R}_{\boldsymbol{v}}(\boldsymbol{\lambda}).$$

Algorithm 2 Globalized semismooth Newton

Input: $\boldsymbol{v}, \gamma, c, L_{\mathrm{d}}$, and tolerance ε.

Set $\boldsymbol{\lambda} = \boldsymbol{0}$.
for $j = 0, 1, \dots$ **do**
  Compute $\boldsymbol{x}(\boldsymbol{\lambda})$, $\mathcal{R}_{\boldsymbol{v}}(\boldsymbol{\lambda})$, $\widehat{\boldsymbol{x}}$, and $\Delta(\widehat{\boldsymbol{x}}, \boldsymbol{\lambda})$.
  if $\Delta(\widehat{\boldsymbol{x}}, \boldsymbol{\lambda}) \le \varepsilon$ **then**
    return $\widehat{\boldsymbol{x}}$.
  end if
  Solve $(\boldsymbol{V} + \sigma_j \boldsymbol{I})\boldsymbol{d} = -\mathcal{R}_{\boldsymbol{v}}(\boldsymbol{\lambda})$ and set $\boldsymbol{p}_{\mathrm{N}} = [\boldsymbol{\lambda} + \boldsymbol{d}]_+ - \boldsymbol{\lambda}$.
  Set $\boldsymbol{p}_{\mathrm{G}} = [\boldsymbol{\lambda} + L_{\mathrm{d}}^{-1} \nabla q_{\boldsymbol{v}}(\boldsymbol{\lambda})]_+ - \boldsymbol{\lambda}$.
  Use $\boldsymbol{p} = \boldsymbol{p}_{\mathrm{N}}$ if it is an ascent direction for $q_{\boldsymbol{v}}$; otherwise use $\boldsymbol{p} = \boldsymbol{p}_{\mathrm{G}}$.
  Choose α by Armijo backtracking on $q_{\boldsymbol{v}}$ and set $\boldsymbol{\lambda} = \boldsymbol{\lambda} + \alpha \boldsymbol{p}$.
end for

Because $\mathcal{R}_{\boldsymbol{v}}$ is piecewise affine, it is strongly semismooth. If the limiting Newton matrices are nonsingular, the regularization vanishes and full steps are eventually accepted, then

$$\|\mathcal{R}_{\boldsymbol{v}}(\boldsymbol{\lambda}^{j+1})\|_2 = \mathcal{O}\left(\|\mathcal{R}_{\boldsymbol{v}}(\boldsymbol{\lambda}^j)\|_2^2\right), \qquad N_{\mathrm{Newton}} = \mathcal{O}\left(\log\log \frac{1}{\varepsilon_{\mathrm{R}}}\right)$$

locally for residual tolerance ε_{R}. After the correct affine cell and active set are identified, one unregularized full Newton step returns the exact proximal point in exact arithmetic, provided $\boldsymbol{V}$ is nonsingular and the step stays in that cell.